1.5 The Energy–Momentum Tensor

By substituting the generalized coordinate q, which represents spacetime, for the field variable φ_a in Eq. (1.4.8), we can rewrite that equation as

$$\frac{\partial \mathcal{L}}{\partial q} - \partial_i \left(\frac{\partial \mathcal{L}}{\partial q_{,i}} \right) = 0 \tag{1.5.1}$$

Here, the Lagrangian density $\mathcal{L}$ can be regarded as a function of the generalized coordinate q and its spacetime derivatives $q_{,i}$, namely, $\mathcal{L} = \mathcal{L}(q, q_{,i})$. The partial derivative of the Lagrangian density with respect to the spacetime coordinate x^i can therefore be expressed as

$$\frac{\partial \mathcal{L}}{\partial x^i} = \frac{\partial \mathcal{L}}{\partial q} \frac{\partial q}{\partial x^i} + \frac{\partial \mathcal{L}}{\partial q_{,k}} \frac{\partial q_{,k}}{\partial x^i} \tag{1.5.2}$$

Substituting Eq. (1.5.1) into the first term on the right-hand side of Eq. (1.5.2), we obtain

$$\frac{\partial \mathcal{L}}{\partial x^i} = \frac{\partial}{\partial x^k} \left(\frac{\partial \mathcal{L}}{\partial q_{,k}} \right) q_{,i} + \frac{\partial \mathcal{L}}{\partial q_{,k}} \frac{\partial q_{,k}}{\partial x^i} \tag{1.5.3}$$

Using the commutativity of mixed partial derivatives, $q_{,k,i} = q_{,i,k}$, we have

$$\frac{\partial \mathcal{L}}{\partial x^i} = \frac{\partial}{\partial x^k} \left(\frac{\partial \mathcal{L}}{\partial q_{,k}} \right) q_{,i} + \frac{\partial \mathcal{L}}{\partial q_{,k}} \frac{\partial q_{,i}}{\partial x^k} = \frac{\partial}{\partial x^k} \left(\frac{\partial \mathcal{L}}{\partial q_{,k}} \cdot q_{,i} \right) \tag{1.5.4}$$

Rearranging Eq. (1.5.4), we obtain the following conservation equation by introducing a new tensor T_i^k:

$$\frac{\partial}{\partial x^k} \left(\frac{\partial \mathcal{L}}{\partial q_{,k}} \cdot q_{,i} - \mathcal{L} \cdot \delta_i^k \right) = \frac{\partial}{\partial x^k} T_i^k = 0 \tag{1.5.5}$$

Here, we introduce an important observation: when the four-dimensional divergence of a four-dimensional current density vector vanishes, the corresponding density satisfies a continuity equation. The continuity equation expresses the conservation of a physical quantity in a closed system: matter cannot be created or destroyed spontaneously. Its physical meaning is that the amount of matter lost within a given spatial region is equal to the amount flowing out of that region. Mathematically, this can be expressed as

$$\frac{\partial}{\partial t}\int \rho\, dV = -\oint \rho \boldsymbol{v} d\boldsymbol{f} \tag{1.5.6}$$

Here, ρ denotes a density, which may represent either electric charge density or energy (mass) density. We first define a four-dimensional current density vector as

$$j^i = (\rho c, \rho \boldsymbol{v}) \tag{1.5.7}$$

We now establish the necessity of the relation, that is, we derive the vanishing four-dimensional divergence of j^i from the continuity equation. Substituting Eq. (1.5.7) into Eq. (1.5.6), we obtain

$$\frac{\partial}{\partial t}\int \rho\, dV + \oint \rho \boldsymbol{v} d\boldsymbol{f} = \frac{\partial}{c\partial t}\int j^0\, dV + \oint j^\alpha df_\alpha = 0 \tag{1.5.8}$$

Integrating Eq. (1.5.8) with respect to the time coordinate and applying the inverse form of Gauss's divergence theorem, we obtain

$$\begin{aligned}\int cdt\left(\frac{\partial}{c\partial t}\int j^0\, dV + \oint j^\alpha df_\alpha\right) &= \int cdt\left(\frac{\partial}{c\partial t}\int j^0\, dV + \int \frac{\partial j^\alpha}{\partial x^\alpha} dV\right) \\ &= \int d\Omega\left(\frac{\partial j^0}{\partial x^0} + \frac{\partial j^\alpha}{\partial x^\alpha}\right) = \int \frac{\partial j^i}{\partial x^i} d\Omega = \int 0 \times cdt = 0\end{aligned} \tag{1.5.9}$$

Since the four-dimensional integration region Ω is arbitrary, we conclude that

$$\frac{\partial j^i}{\partial x^i} = 0 \tag{1.5.10}$$

We now establish the sufficiency of the relation, that is, we derive the continuity equation (1.5.6) from the vanishing four-dimensional divergence of j^i. Integrating Eq. (1.5.10) over the entire spacetime and applying Gauss's divergence theorem, we obtain

$$\int \frac{\partial j^i}{\partial x^i} d\Omega = \oint j^i dS_i \tag{1.5.11}$$

For Eq. (1.5.11), we can construct the decomposition illustrated in Fig.1.5.

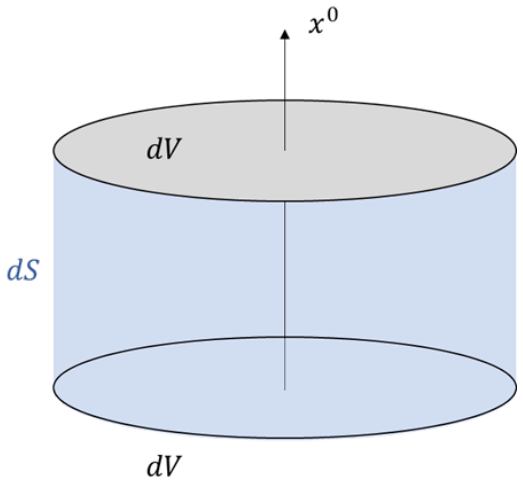


Fig.1.5. Illustration of the four-dimensional divergence

According to Fig.1.5, we decompose the hypersurface integral in Eq. (1.5.11) into three contributions: the upper hypersurface $dS_{01} = dV$, the lower hypersurface $dS_{02} = dV$, and the surrounding hypersurface dS. The integrals over the upper and lower hypersurfaces correspond to spatial volume integrals at two different times, respectively, with both hypersurfaces perpendicular to the time axis x^0. The surrounding hypersurface dS extends to spatial infinity. Since no matter or fields are present at infinity, the integrals of any tensor, vector, or scalar quantity over this hypersurface vanish. Consequently, Eq. (1.5.11) reduces to

$$\begin{aligned} \oint j^i dS_i &= \int 0 dS + \int j^0 dS_{01} + \int j^0 dS_{02} \\ &= \int j^0(x^0 + dx^0) dV - \int j^0(x^0) dV \end{aligned} \tag{1.5.12}$$

Substituting Eq. (1.5.10) into Eq. (1.5.12), we obtain

$$\int j^0(x^0 + dx^0) dV = \int j^0(x^0) dV \tag{1.5.13}$$

This equation indicates that the total amount of the conserved quantity over the entire spatial volume remains unchanged with time. Consequently, within any local spatial region, the amount of the quantity lost must equal the amount gained by the surrounding region. This is precisely the continuity equation.

In summary, when the four-dimensional divergence of a four-vector or tensor vanishes, the physical quantity represented by that vector or tensor satisfies a conservation law in a closed system. According to Eq. (1.5.5), the newly defined tensor T_i^k satisfies this condition. We therefore identify this tensor as the **energy-momentum tensor**, whose explicit form will be derived later in the context of specific field theories.